

Design and Development of a Tape-Type Vibrotactile Feedback System

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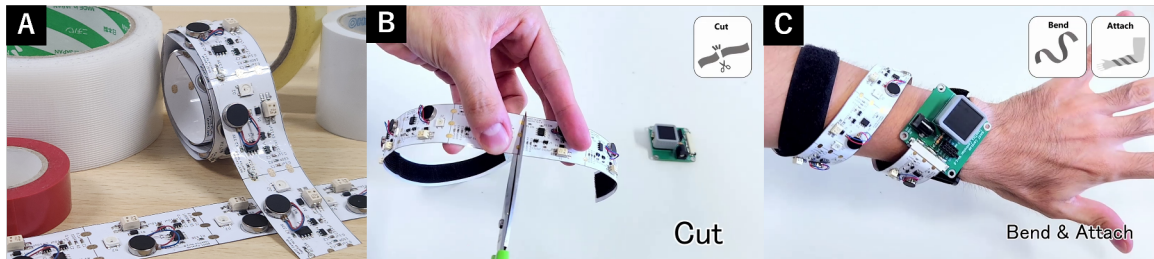


Figure 1: (A) Appearance of Tape-Tics; (B) Cutting Tape-Tics; (C) Wrapping Tape-Tics around an arm

ABSTRACT

Tape-Tics is a flexible, modular vibrotactile feedback system integrating LEDs and vibration motors. It is user-friendly, requiring no specialized knowledge, and is easily controlled via a PC-based GUI. Users can set vibration intensity, LED colors, and transmit configurations using Bluetooth Low Energy (BLE), enabling operation based on predefined settings. Built with flexible printed circuit (FPC) technology, Tape-Tics adapts seamlessly to curved surfaces, enhancing its versatility for educational, research, and artistic applications.

In a hands-on workshop, high school students engaged in creative projects using Tape-Tics, assembling and operating the device to explore vibrotactile technology. Through the GUI controller, they configured vibration patterns and developed unique applications. Surveys revealed that participants' understanding of haptics improved significantly, and they rated Tape-Tics highly as an educational tool.

Index Terms: Haptic interfaces, Flexible printed circuits, Rapid prototyping, Wearable devices, Educational technology.

1 INTRODUCTION

As VR becomes increasingly popular, tactile feedback has gained attention for enhancing immersive experiences. Technologies such as linear resonant actuators (LRA), and eccentric rotating mass (ERM) motors are commonly used in devices like gaming consoles and mobile phones. However, when prototyping or developing devices that utilize a large number of vibrators, complex wiring becomes a significant challenge.

While toolkits, such as the TECHTILE Toolkit [5] offer frameworks for haptic feedback, they often require significant technical expertise for expansion. Flexible devices such as SensorTape [2] are designed with pre-mounted components on flexible substrates,

making them easier to use without complex assembly. Reconfigurable modular haptic devices including SoftVibe [4] and VibroForge [3] allow users to combine and customize the shape, size, and type of tactile stimulation easily. Modular devices like ChainFORM [6] provide flexibility but are less accessible due to intricate assembly requirements.

To address these challenges, we propose Tape-Tics, a modular, tape-like device inspired by LED strip lights. Tape-Tics features small vibrators integrated with NeoPixel-controlled LEDs, mounted on a flexible printed circuit (FPC) board. Users can cut Tape-Tics to any desired length, attach it to various surfaces, and receive both visual and tactile feedback. Each node's vibrator and LED can be independently controlled, with vibration intensity adjustable in 256 levels. Leveraging the NeoPixel library ensures compatibility with extensive online resources, enhancing scalability.

To simplify usage, we provided a GUI controller that runs on a PC, enabling users to adjust settings and operate the device via Bluetooth without advanced programming skills. This user-friendly approach lowers the barrier for non-experts.

We hosted a hands-on workshop for high school students, who frequently engage in creative projects, to test the effectiveness of the device. After explanation of how to use Tape-Tics, participants were encouraged to create projects using the device. As a result, all participants were able to use the device with ease, enthusiastically modifying and attaching it to their bodies or other objects. The unique projects they created, such as a vibration alarm wristband and a hat that notifies the wearer through vibrations, demonstrated their growing interest in vibrotactile technology. Survey results indicated that Tape-Tics was perceived as an effective tool for incorporating vibrotactile feedback into education.

The main contributions are summarized as follows:

1. Proposing a novel design for a haptic device that lowers the difficulty of development, making it accessible to users without specialized knowledge, and integrating a user-friendly GUI controller that runs on PC to ensure easy operation for beginners.
2. Reviewing the price and power consumption to verify if the device can be introduced to educational institutions, and specifying the maximum length.
3. Confirming the effectiveness of the device through workshops and reviewing the applications created by the participants.

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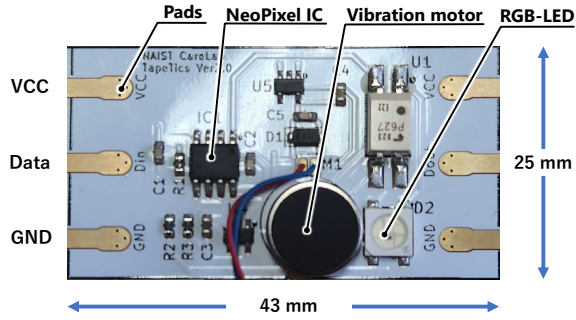


Figure 2: Components of the Tape Node

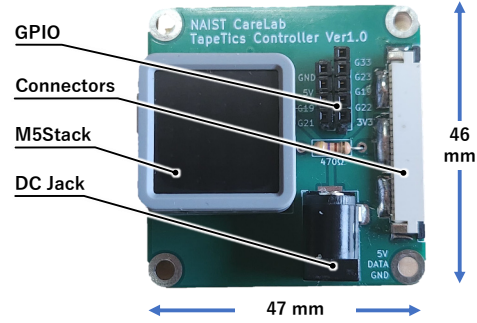


Figure 3: Components of the Master Node

2 DESIGN AND IMPLEMENTATIONS

2.1 Hardware

The hardware is designed to allow each node to control both the vibration intensity and LED brightness independently. The control unit, equipped with a microcontroller, integrates easily with a variety of applications, providing users with a customizable experience.

2.1.1 Tape Node

The configuration of the Tape Node is shown in Figure 2. Considering the need to attach the device to curved surfaces, it was developed using flexible printed circuit (FPC) technology. The substrate thickness is 0.11 mm, and a 2-layer polyimide-based substrate was used. Nodes are arranged with a 40 mm spacing, and the tape width is 25 mm to ensure compatibility with the FFC connector on the Master Node. Power is supplied from a 5V source, with two voltage regulators providing 5V to the WS2811 IC and LEDs, and 3.3V to the vibrator. Additionally, a signal is sent from the WS2811 to the vibrator, and a photocoupler is used to isolate the WS2811 circuit from the vibrator. To allow for flexibility, larger components were placed as centrally as possible. There are three solder pads on both sides of the tape: power, data, and ground lines, allowing for easy connections. The circuit diagram of the Tape Node is shown in Appendix A of the Additional Materials.

2.1.2 Master Node

The configuration of the Master Node is shown in Figure 3. It features a controller unit with dimensions of 47 mm in height and 46 mm in width, equipped with an M5Stack Atom Matrix. Wireless communication with the computer is performed via Bluetooth Low Energy. The unit includes GPIO pins that can be combined with other sensors. The drive unit is connected via an FFC connector for flexible PCB, and power is supplied from an external battery providing DC5V. The circuit diagram of the Master Node is shown in Appendix B of the Additional Materials.

2.2 Software

2.2.1 Communication

The WS2811 chips communicate with each other through serial data transmission. When sending data from a PC, the first LED chip receives its own RGB value data, shapes the remaining data, and then sends the color data to the next WS2811 chip. This eliminates the need for complex wiring and the individual slave and address settings required in I2C (Inter-Integrated Circuit) communication. The communication speed between WS2811 chips is 800Kbps, which is faster than the 100Kbps to 400Kbps speeds typically seen in I2C communication¹.

¹<https://www.nxp.com/docs/en/user-guide/UM10204.pdf>

2.2.2 LED Control Library

NeoPixel is a brand by Adafruit for their WS2812 LEDs and compatible products². NeoPixel (WS2812) color LEDs come with an integrated IC that controls the LED's color and brightness using a single data line. This setup allows for chaining multiple color LEDs together while still being controlled by a single signal line. To generate the control signal for these LEDs, high-speed and precise timing is required. Programming for this can be complex, requiring knowledge of MCU assembly language. However, Adafruit offers an open-source NeoPixel library that simplifies the control of NeoPixel (WS2812) LEDs. The NeoPixel series operates with just three connections: power, signal line, and ground (GND). The NeoPixel library works with platforms like Arduino, ESP32, M5Stack, and Raspberry Pi. In this research, we opted for FastLED, which is faster and more memory-efficient compared to NeoPixel, and is suitable for use within the limited memory of M5Stack.

2.2.3 GUI Controller

The development of Tape-Tics can be controlled using a graphical user interface (GUI) running on a PC, created with Max8 (Cycling74)³. The appearance of the GUI is shown in Figure 4. No keyboard input is required, as users can operate the interface entirely through drag-adjustable sliders and button clicks. At the top, there are sliders to set the drive time and maximum vibration intensity (Figure 4 (a)). Pressing the "Set" button transfers the configuration to a Python program, where it is formatted into a structure readable by Tape-Tics (Figure 4 (b)). The "Send" button transmits the information to Tape-Tics via BLE communication, and pressing the "Run" button starts the operation based on the predefined settings. At the center, there is a timeline panel commonly seen in video editing software (Figure 4 (c)), with the horizontal axis representing time and the vertical axis representing vibration intensity. By clicking directly on the timeline, users can set the timing and intensity of the vibrations. The node number can be adjusted using the up and down buttons on the left side, allowing configurations for each individual node. At the bottom, there are radio buttons for setting the LED color (Red, Green, or Blue) for each node (Figure 4 (d)). Additionally, settings can be saved to a text file and reloaded for future use, making it convenient for repeated sessions. This design allows non-programmers to easily control Tape-Tics without requiring any coding skills.

3 DEVICE COST AND LENGTH

The manufacturing cost of a Tape Node is approximately \$2.697 USD per unit. For the Master Node, the total cost per PCB is \$21.95

²https://github.com/adafruit/Adafruit_NeoPixel

³<https://cycling74.com/>

Table 1: Results of the questionnaire

Question No.	Questionnaire	Mean	Median	SD
Pre-workshop survey				
Q1	How much do you know about haptic technology?	2.3	2.0	1.160
Q2	Are you aware of haptic technology in your daily life?	3.6	4.0	1.075
Post-workshop survey				
Q3	Did you understand the tactile lecture?	4.1	4.5	1.287
Q4	Would you like to pursue haptic design in the future?	4.2	4.0	0.632
Q5	How effective was Tape-Tics as an educational tool?	4.6	5.0	0.699
Q6	How much did you enjoy the workshop?	4.9	5.0	0.316
Q7	Was the controller operation easy to understand?	4.4	4.5	0.699
Q8	How easy was the device to flex, cut, and install?	4.2	4.0	0.632
Q9	Was it easy to insert and remove the tape to and from the connector?	3.7	4.0	0.949
Q10	How much of the work is complete?	3.7	4.0	0.675
Q11	Did the implementation go smoothly?	3.8	4.0	0.632

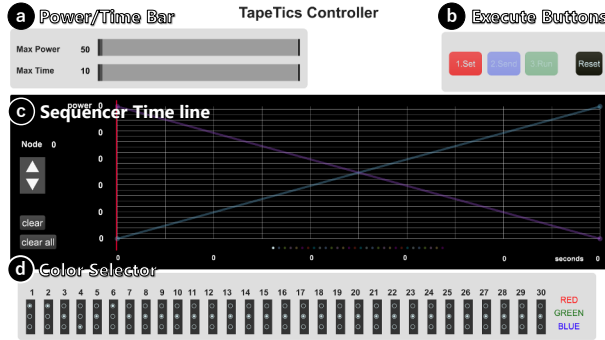


Figure 4: User Interface of the Tape-Tics GUI Controller: (a) Slider to control the time and the maximum vibration power; (b) Buttons to execute the sequence; (c) Timeline to control the vibration for each node; (d) Radio buttons to select the color for each node.

USD. The power consumption per Tape Node is 128 mA, which includes the vibrator (80 mA) and the LED (48 mA). The M5Stack controller consumes 80 mA (0.4 W).

Using a copper plane with a maximum current rating of 10 A, up to 78 nodes can be operated simultaneously. With a mobile battery rated at 3.6 V, 3350 mAh, and an output current of 2.4 A, the maximum number of nodes is 18, providing an operational runtime of approximately 1.39 hours.

4 USER STUDY: TAPE-TICS WORKSHOP

4.1 User Evaluation

To evaluate the usability of Tape-Tics for individuals without specialized knowledge, we conducted a hands-on workshop. The aim was to observe whether participants understood how to use Tape-Tics and to explore the creative works they could produce. Ethical approval for this study was obtained from the Institutional Review Board of our organization. The workshop included 10 male students aged 15–18 ($M = 16.3$, $SD = 0.9$), comprising two Mechanics majors and eight Electronics majors. The workshop was held in the computer lab at the high school. It lasted two days, each approximately six hours long.

The analysis included quantitative data from surveys and qualitative insights from interviews, sticky notes, presentation materials, recorded videos, audio recordings, and open-ended survey responses. Additionally, key themes were identified during the analysis [1]. Evaluations were conducted using a 5-point Likert scale,

where 1 indicated “strongly disagree” and 5 indicated “strongly agree.”

5 ANALYSIS AND RESULTS

5.1 Quantitative analysis

In the pre-workshop surveys, the mean score for Q1, “How much do you know about haptic technology?” was 2.3, indicating that participants’ prior knowledge was generally low. In contrast, post-workshop responses showed an increase in understanding, as reflected in Q3, “Did you understand the tactile lecture?” which received a mean score of 4.1.

Additionally, participants expressed high satisfaction with the overall workshop experience, as shown in Q6, “How much did you enjoy the workshop?” which received a mean score of 4.9, indicating that most participants found the workshop enjoyable. Most participants found the workshop engaging, particularly the hands-on activities and teamwork. They enjoyed exploring Tape-Tics’ flexibility and creating unique designs.

Regarding the usability of the Tape-Tics system, Q7, “Was the controller operation easy to understand?” received a mean score of 4.4, indicating that the controller was generally easy to use. The flexibility and ease of installation of the device, as assessed in Q8, “How easy was the device to flex, cut, and install?” received a positive rating, with a mean score of 4.2. However, inserting and removing tape from the connector received a lower score of 3.7 in Q9, “Was it easy to insert and remove the tape from the connector?” suggesting that this area requires improvement.

Project completion and implementation smoothness were rated moderately, with mean scores of 3.7 for Q10, “How much of the work is complete?” and 3.8 for Q11, “Did the implementation go smoothly?”, reflecting some technical challenges during the project. Participants suggested several improvements, including more precise frequency and color customization, waveform import/export features, and adjustments to the size and weight of the device for better usability.

5.2 Qualitative analysis

After analyzing the post-workshop data, three major themes emerged from the participants’ feedback: Understanding of Haptic Technology, Enjoyment and Satisfaction with the Workshop, and Improvements to the Controller and Device.

Understanding of haptic technology - Many participants expressed that they made new discoveries about haptic technology and its applications through the workshop. In particular, they mentioned that they realized how vibration can serve various functions

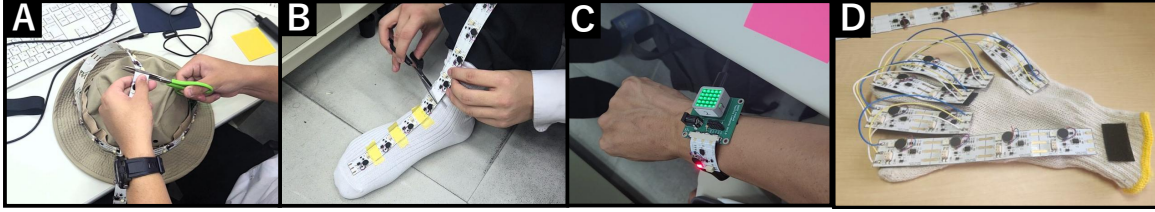


Figure 5: Scenes of project creation: (A) Cutting Tape-Tics; (B) Attaching Tape-Tics to a foot; (C) Vibration alarm wristband created by a participant; (D) Example of a haptic glove made by attaching Tape-Tics to a work glove;

and help people perceive things they might not normally notice or interact with their bodies.

I was surprised that vibration alone can create such convenient functions. - (P1, male, 16 years)

I realized that haptics is one of the most primitive senses we possess... and that this technology will undoubtedly be necessary for the future. - (P9, Male, 16 years)

Enjoyment and satisfaction of the workshop - Many teams created unique designs by utilizing the flexibility of Tape-Tics, but several areas needing technical improvement were also highlighted. In addition to feedback on the direction of tape attachment and adjustments to the PCB size, there were requests to strengthen the vibration and expand the color variations of the LEDs. Based on this feedback, it is clear that further technical improvements will be necessary for future projects. Many participants enjoyed working in teams, sharing ideas, and creating projects, with the trial-and-error aspect of using Tape-Tics as a source of enjoyment.

I had a lot of fun when programming and trying different things. - (P2, Male, 17 years)

It was enjoyable to work together and come up with ideas, and to have fun as a group. - (P3, male, 15 years)

Improvement of the controller and device - Many teams, as seen in Figure 5, cut the tape with scissors, attached it to hats or bags, and wrapped it around their arms. One team, aiming to create a shoe-shaped device, soldered Tape-Tics and cables to accommodate the complex shape.

Many participants also provided specific suggestions for improving the controller and device. In particular, they requested features such as the ability to move points freely, more detailed customization of frequency and color, and the addition of waveform import/export functions. Feedback was also received regarding the size and weight of the device and controller. While some participants were satisfied with the current size, others found it slightly heavy and offered specific suggestions for the ideal size.

I would like to have the ability to adjust the frequency and specify RGB colors for the LED since it supports full color, along with brightness control. - (P4, Male, 15 years)

It would be great to have a waveform import and export function. - (P7, Male, 17 years)

I think the current size is fine, but it feels a little heavy. - (P2, Male, 17 years)

The Controller Node size should be about the area of a 30mm by 30mm. - (P1, Male, 16 years)

The Tape Node should be more smaller. - (P8, Male, 16 years)

6 CONCLUSION AND FUTURE WORK

In conclusion, Tape-Tics offers a versatile, flexible, and user-friendly system for providing vibrotactile feedback. The hands-on workshop demonstrated its effectiveness in teaching haptic concepts and fostering creative applications, such as VR haptic gloves

and wristwatch-type devices(5 (c) (d)). The positive feedback and success of the workshop confirm Tape-Tics' potential as both an educational tool and a research device.

The complexity of integrating additional features, such as sensors, was a challenge due to limited GPIO pins. Future versions will include plug-and-play sensors compatible with the Grove system, which will simplify sensor integration. Additionally, feedback indicated that vibration strength was insufficient for some applications. Future iterations may incorporate more powerful motors, such as LRA, to address this.

While the current GUI is effective for beginners, greater platform compatibility, particularly with smartphones, will be prioritized to improve accessibility. The issue of incorrect connector insertion will be addressed by adding visual guides. Challenges with attaching Tape-Tics modules to objects or body parts will be resolved by refining adhesive properties or developing specialized attachments.

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